

Trainings ▾

General Information

⚠ WARNING ⚠

Fuel is flammable. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas sharing ventilation to reduce the possibility of severe personal injury or death when working on the fuel system.

⚠ WARNING ⚠

Do not vent the fuel system on a hot engine; this can cause fuel to spill onto a hot exhaust manifold, which can cause a fire.

The low-pressure fuel system for a Cummins® diesel installed in the vehicle consists of the fuel tank, lines between the tank and the engine, transfer pump and lines, and fuel filter and lines. Air or bubbles at the injection pump can cause the engine **not** to operate or erratic engine operation and/or subsequent malfunction of the fuel injection pump. Air can be introduced by leaks in the fuel system prior to the transfer pump since the fuel in the lines is under a vacuum and can draw air into the system. Bubbles can result from any number of restrictions in the system:

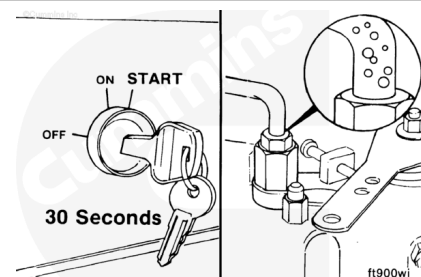
- Plugged fuel filter
- Crimped fuel line.

If sufficient fuel reaches the injection pump from the low-pressure system then solutions to engine operational problems are elsewhere. The following steps will aid in evaluating low-pressure fuel system performance.

Replacing the fuel supply lines, fuel filters, fuel injection pump, high-pressure fuel lines, and injectors will let air enter into the fuel system. Follow the specified procedure to bleed the air from the system.

Use the following procedure for proper venting of the low pressure side of the fuel system. Refer to Procedure 006-015 (</qs3/pubsys2/xml/en/procedures/40/40-006-015-tr.html>) in Section 6.

Use the following procedure to determine if venting the fuel pump is necessary. Refer to Procedure 005-012 (</qs3/pubsys2/xml/en/procedures/40/40-005-012-tr.html>), or

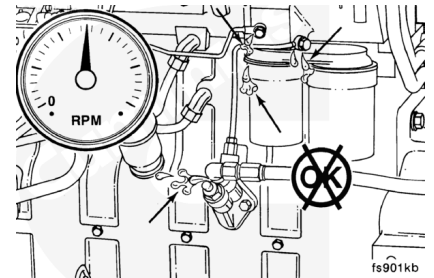


Refer to Procedure 005-014

(/qs3/pubsys2/xml/en/procedures/40/40-005-014-tr.html) in Section 5.

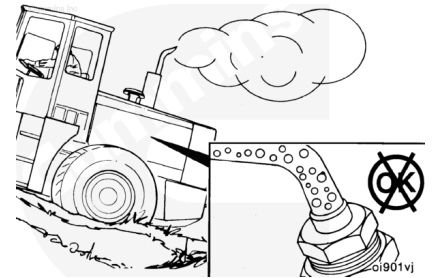
Use the following procedure for venting of the high pressure side of the fuel system. Refer to Procedure 006-051 (/qs3/pubsys2/xml/en/procedures/40/40-006-051-tr.html) in Section 6.

Since the fuel transfer pump provides positive pressure through the fuel filter and supply line to the fuel injection pump, loose connections or defective seals show as a fuel leak and **not** as an air leak.



Air in the fuel system will cause the following engine problems:

- Hard to start
- Run rough
- Misfire
- Fuel knock.



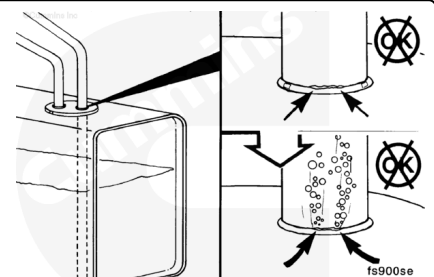
Note : If an excessive amount of air has entered into the system, the system will need to be vented. Follow the “Prime” sections referenced above.

Test

A source that is often overlooked for air to enter the fuel system is between the inlet of the prefilter and the suction tube in the tank. Fuel tanks that have the outlet fitting at the top have a suction tube that extends to the bottom of the tank. Cracks or pin holes in the weld that join the tube to the fitting can let air enter the fuel system.

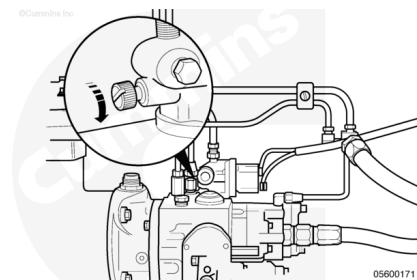
Also, check to make sure all the fittings from the fuel supply line on the tank to the inlet of the fuel transfer pump are tight.

Use a sight glass at the fuel lift pump inlet to check for air in the fuel supply lines.



Before venting the air in the fuel, turn the knob on the shutoff valve **clockwise** to open the valve.

After finish venting, turn the knob on the shutoff valve **counterclockwise** to close the valve.



A stuck-open injector can also blow combustion gas back into the pump and cause air to be present in the overflow. If the engine seems to be misfiring or running rough, break all the injector supply lines loose at the pump end.

Note : The injector supply lines must be loosened, and tightened, one at a time until the issue is resolved.

Note : Use two wrenches when loosening the lines at the fuel pump, one to hold the delivery valve and one to loosen the fuel line.

Tighten the fuel lines.

Torque Value: 24 n•m [212 in-lb]

